

Specifications of 2-Phase Stepping Motor

General Specifications

	103-4902	103-591	103-771□	103H32□□/SH35□□/103H52□□
Insulation class	Class B (130°C)			
Insulation resistance	Not less than 100MΩ between winding and frame by DC500V megger or normal temperture and humidity.			
Withstand voltage	Without abnormality when applying 50/60Hz, 0.5kV AC (1kV AC for 103-771□) for 1minute (leakage current 1mA) between winding and frame at nomal temperature and humidity.			
Operating environment	Ambient temperature: -10°C~+50°C			
	Ambient humidity: 20~90% RH (no condensation)			
Winding temperature rise	80K MAX. (Based on Sanyo Denki standard)			
Standing angle error	±0.045°	±0.054°	±0.045°	±0.09°
Axial play	0.075mm(0.003inch) MAX	0.075mm(0.003inch) MAX	0.075mm(0.003inch) MAX	0.075mm(0.003inch) MAX
	Load 4.4N(1lbs)	Load 4.4N(1lbs)	Load 9N(2lbs)	Load 4.4N(1lbs)
Radial play (Note 1)	0.025mm(0.001inch) MAX Load 4.4N(1lbs)			
Shaft runouts	0.025mm(0.001inch)			
Concentricity of mounting spigot relative to shaft	ø0.05mm(0.002inch)	ø0.05mm(0.002inch)	ø0.075mm(0.003inch)	ø0.05mm(0.002inch)
Perpendicularity of mounting surface relative to shaft	0.075mm(0.03inch)	0.075mm(0.03inch)	0.075mm(0.003inch)	0.1mm(0.004inch)

(Note 1) When load is applied at 1/3 length from output shaft edge.

	103H670□	103H712□/103H782□	103H822□□/103H8922□□
Insulation class	Class B (130°C)		
Insulation resistance	Not less than 100MΩ between winding and frame by DC500V megger or normal temperture and humidity.		
Withstand voltage	Without abnormality when applying 50/60Hz, 1kV AC for 1minute (leakage current 1mA) between winding and frame at nomal temperature and humidity.		
Operating environment	Ambient temperature: -10°C~+50°C		
	Ambient humidity: 20~90% RH (no condensation)		
Winding temperature rise	80K MAX. (Based on Sanyo Denki standard)		
Standing angle error	±0.09°	±0.054°	±0.09°
Axial play	0.075mm(0.003inch) MAX. Load 9N(2lbs)		
Radial play (Note 1)	0.025mm MAX. Load 4.4N(1lbs)		
Shaft runouts	0.025mm(0.001inch)		
Concentricity of mounting spigot relative to shaft	ø0.075mm(0.003inch)		
Perpendicularity of mounting surface relative to shaft	0.075mm(0.003inch)		

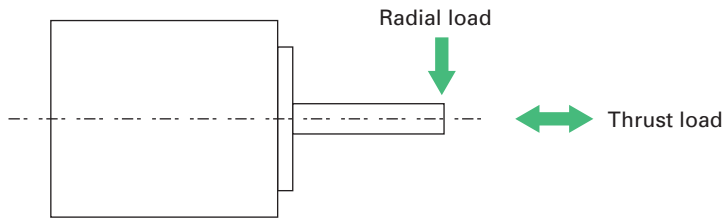
(Note 1) When load is applied at 1/3 length from output shaft edge.

General Specifications (CE Marked Models)

	103H712□	103H8922□/103H822□
Rated voltage	12-200VDC	12-300VDC
Applied standards(Low voltage directive)	EN60034-1, IEC34-5(EN60034-5), EN60204-1, EN60950, EN61010-1	
Operation type	S1 (continuous rating)	
Protection grade	IP43	
Device category	Class I	
Operation environment	Pollution dgree	
Insulation classl	Class B (130°C)	
Insulation resistance	Not less than 100MΩ between winding and frame by DC500V megger or normal temperture and humidity.	
Withstand voltage	Without abnormality when applying 50/60Hz, 1600V AC (1500V AC for 103H712□) for 1minute (leakage current 10mA) between winding and frame at nomal temperature and humidity.	
Operating environment	Ambient temperature: -10°C~+40°C	
	Ambient humidity 20~90% (no condensation)	
Winding temperature rise	80K MAX. (Based on Sanyo Denki standard)	
Standing angle error	±0.054°	±0.09°
Axial play	0.075mm(0.003inch) MAX. Load 9N(2lbs)	
Radial play (Note 1)	0.025mm(0.001inch) MAX. Load 4.4N(1lbs)	
Shaft runouts	0.025mm(.001inch)	
Concentricity of mounting spigot relative to shaft	ø0.075mm(0.003inch)	
Perpendicularity of mounting surface relative to shaft	0.1mm(0.04inch)	

(Note 1) When load is applied at 1/3 length from output shaft edge.

Allowable radial load / thrust load



Frange size	Model. No.	Distance from end of shaft:mm (inch)				Thrust load N(lbs)
		0	5(0.20)	10(0.39)	15(0.59)	
		Radial load:N(lbs)				
□28mm(□1.10inch)	103H32□□	30(6)	39(8)	53(11)	84(18)	3(0.67)
□35mm(□1.38inch)	SH35□□	40(8)	50(11)	67(15)	98(22)	10(2.25)
□39mm(□1.54inch)	103-49□□	43(9)	59(13)	93(20)	216(48)	30(6.75)
□42mm(□1.65inch)	103H52□□	22(4)	26(5)	33(7)	46(10)	10(2.25)
	103-59□					
□50mm(□1.97inch)	103H670□	71(15)	87(19)	115(25)	167(37)	15(3.37)
□56mm(□2.20inch)	103H712□	52(11)	65(14)	85(19)	123(27)	15(3.37)
	103H7128	85(19)	105(23)	138(31)	200(44)	15(3.37)
φ 56mm(φ 2.20inch)	103-77□□	75(16)	92(20)	121(27)	176(39)	15(3.37)
□60mm(□2.36inch)	103H782□	70(15)	87(19)	114(25)	165(37)	20(4.50)
φ 86mm(φ 3.39inch)	103H822□	167(37)	193(43)	229(51)	280(62)	60(13.488)
φ 106mm(φ 4.17inch)	103H8922□	321(72)	357(80)	401(90)	458(102)	100(22.48)

39mm(1.54/0.9)
42mm(1.65/0.9)

28mm(1.10/1.8)
35mm(1.38/1.8)

42mm(1.65/1.8)
50mm(1.97/1.8)

56mm(2.20/1.8)
60mm(2.36/1.8)

86mm(3.39/1.8)
106mm(4.17/1.8)

106mm(4.17/1.8)/CE
56mm(2.20/1.8)

86mm(3.39/1.8)
106mm(4.17/1.8)

106mm(4.17/1.8)/CE
56mm(2.20/1.8)

106mm(4.17/1.8)/CE
56mm(2.20/1.8)

106mm(4.17/1.8)/CE
56mm(2.20/1.8)

106mm(4.17/1.8)/CE
56mm(2.20/1.8)

106mm(4.17/1.8)/CE
56mm(2.20/1.8)

106mm(4.17/1.8)/CE
56mm(2.20/1.8)

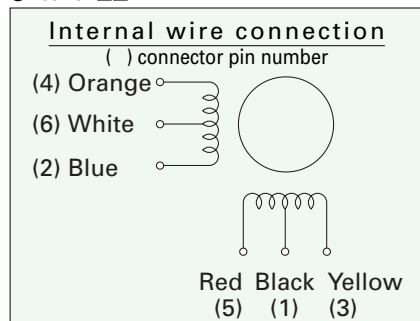
106mm(4.17/1.8)/CE
56mm(2.20/1.8)

106mm(4.17/1.8)/CE
56mm(2.20/1.8)

Internal Wiring and Rotation Direction

Unipolar winding

● 103H32□□

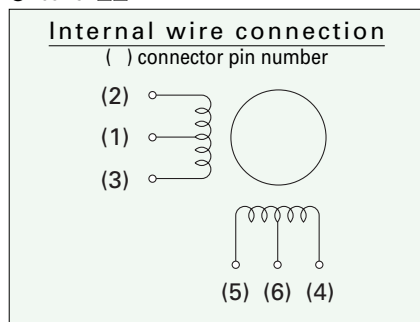


Direction of motor rotate

The output shaft shall rotate clockwise as seen from the shaft side, when excited by DC in the following order.

		Lead wire color, connector type pin number				
Lead wire		White & black	Red	Blue	Yellow	Orange
Connector		(1.6)	(5)	(2)	(3)	(4)
Exciting order	1	+	-	-	-	-
	2	+	-	-	-	-
	3	+	-	-	-	-
	4	+	-	-	-	-

● 103H52□□

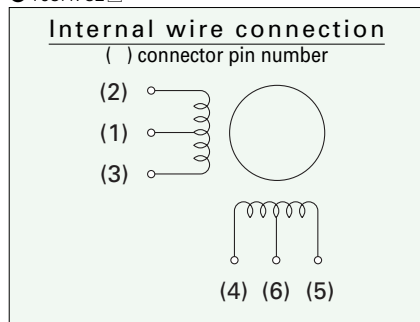


Direction of motor rotate

The output shaft shall rotate clockwise as seen from the shaft side, when excited by DC in the following order.

		Connector type pin number				
		(1.6)	(5)	(3)	(4)	(2)
Exciting order	1	+	-	-	-	-
	2	+	-	-	-	-
	3	+	-	-	-	-
	4	+	-	-	-	-

● 103H782□

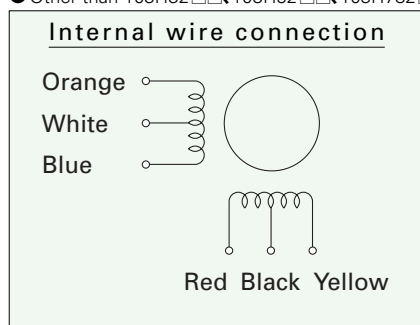


Direction of motor rotate

The output shaft shall rotate clockwise as seen from the shaft side, when excited by DC in the following order.

		Connector type pin number				
		(1.6)	(4)	(3)	(5)	(2)
Exciting order	1	+	-	-	-	-
	2	+	-	-	-	-
	3	+	-	-	-	-
	4	+	-	-	-	-

● Other than 103H32□□, 103H52□□, 103H782□

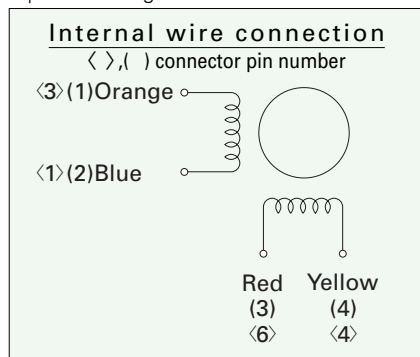


Direction of motor rotate

The output shaft shall rotate clockwise as seen from the shaft side, when excited by DC in the following order.

		Lead wire color				
Lead wire		White & black	Red	Blue	Yellow	Orange
Exciting order	1	+	-	-	-	-
	2	+	-	-	-	-
	3	+	-	-	-	-
	4	+	-	-	-	-

Bipolar winding



Direction of motor rotate

The output shaft shall rotate clockwise as seen from the shaft side, when excited by DC in the following order.

		Lead wire color, connector type pin number			
Lead wire		Red	Blue	Yellow	Orange
Exciting order	1	-	-	+	+
	2	+	-	-	+
	3	+	+	-	-
	4	-	+	+	-
Connector	103H782□	(3)	(2)	(4)	(1)



2-phase Stepping Motor

86mm cir. 103H822□
(3.39inch cir.) 1.8°/step

Recommendable Driver
Refer to the page 7,17,27 and 45.

Specifications

Unipolar winding

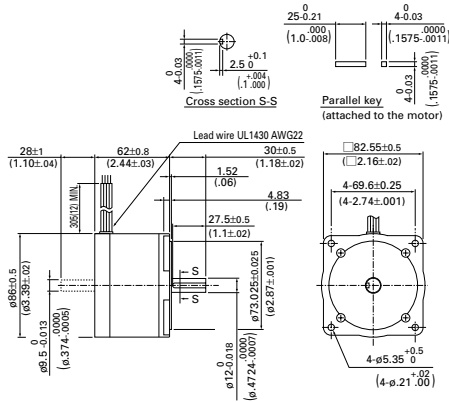
Model		Holding torque at 2-phase energization	Rated current	Resistance	Inductance	Rotor inertia	Mass(Weight)
Single shaft	Double shaft	N·m (oz·in) MIN.	A/phase	Ω/phase	mH/phase	$\times 10^{-4}$ kg·m ² (oz·in ²)	kg (lbs)
103H8221-0441	-0411	2.15(304.5)	2	2.5	7.2	1.45(7.93)	1.5(3.31)
103H8221-0941	-0911	2.15(304.5)	4	0.62	1.8	1.45(7.93)	1.5(3.31)
103H8222-0441	-0411	4.13(584.8)	2	4.0	15	2.9(15.86)	2.5(5.51)
103H8222-0941	-0911	4.13(584.8)	4	0.97	3.6	2.9(15.86)	2.5(5.51)
103H8223-0441	-0411	6.27(887.9)	2	5.6	24	4.4(24.06)	3.5(7.72)
103H8223-0941	-0911	6.27(887.9)	4	1.35	5.6	4.4(24.06)	3.5(7.72)

Bipolar winding

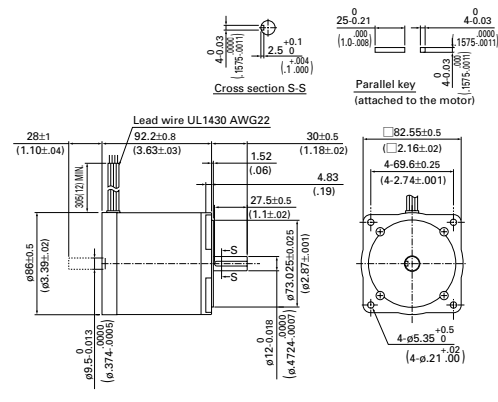
Model		Holding torque at 2-phase energization	Rated current	Resistance	Inductance	Rotor inertia	Mass(Weight)
Single shaft	Double shaft	N·m (oz·in) MIN.	A/phase	Ω/phase	mH/phase	$\times 10^{-4}$ kg·m ² (oz·in ²)	kg (lbs)
103H8221-5041	-5011	2.74(388.0)	2	2.3	14	1.45(7.93)	1.5(3.31)
103H8221-5141	-5111	2.74(388.0)	4	0.6	3.5	1.45(7.93)	1.5(3.31)
103H8221-5241	-5211	2.74(388.0)	6	0.3	1.65	1.45(7.93)	1.5(3.31)
103H8222-5041	-5011	5.09(720.8)	2	2.7	23	2.9(15.86)	2.5(5.51)
103H8222-5141	-5111	5.09(720.8)	4	0.7	5.7	2.9(15.86)	2.5(5.51)
103H8222-5241	-5211	5.09(720.8)	6	0.35	2.7	2.9(15.86)	2.5(5.51)
103H8223-5041	-5011	7.44(1053.6)	2	3.6	32.5	4.4(24.06)	3.5(7.72)
103H8223-5141	-5111	7.44(1053.6)	4	0.9	8.1	4.4(24.06)	3.5(7.72)
103H8223-5241	-5211	7.44(1053.6)	6	0.45	3.4	4.4(24.06)	3.5(7.72)

Dimensions [Unit:mm(inch)]

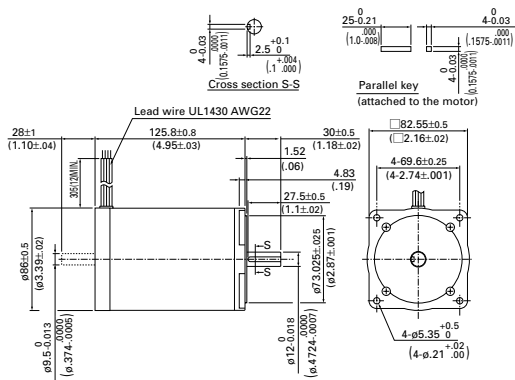
103H8221-0441/0941 (Single shaft)
103H8221-0411/0911 (Double shaft)



103H8222-0441/0941 (Single shaft)
103H8222-0411/0911 (Double shaft)

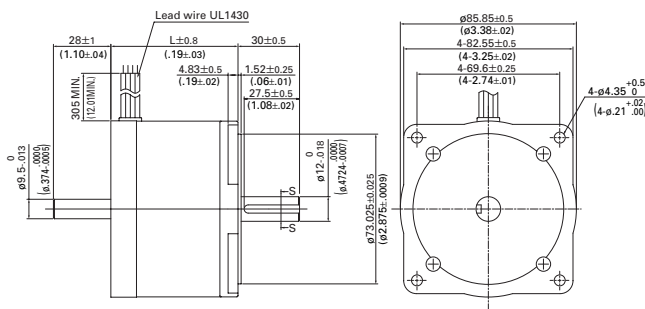


103H8223-0441/0941 (Single shaft)
103H8223-0411/0911 (Double shaft)



Bipolar winding

103H8221/103H8222/103H8223-5 □ 41 (Single shaft)
103H8221/103H8222/103H8223-5 □ 11 (Double shaft)

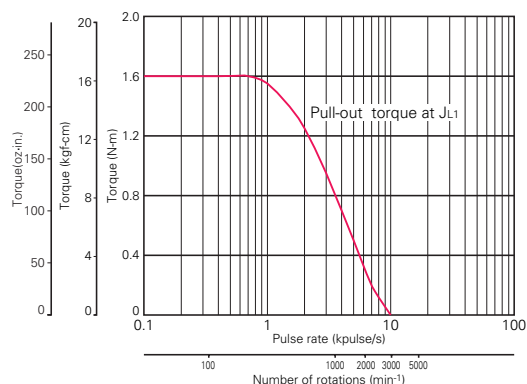


Model	AWG	L
103H8221-50 □ 1	22	62 (2.44)
103H8221-51 □ 1		
103H8221-52 □ 1		
103H8222-50 □ 1	22	92.2 (3.63)
103H8222-51 □ 1		
103H8222-52 □ 1		
103H8223-50 □ 1	22	125.8 (4.95)
103H8223-51 □ 1		
103H8223-52 □ 1		

35mm(1.38)/1.8"
38mm(1.54)/0.9"
42mm(1.65)/0.9"
28mm(1.10)/1.8"
42mm(1.65)/1.8"
50mm(1.97)/1.8"
56mm(2.20)/1.8"
60mm(2.36)/1.8"
86mm(3.39)/1.8"
106mm(4.17)/1.8"
56mm(2.20)/CE
86mm(3.39)/CE
106mm(4.17)/CE
Specifications of 2-phase stepping motor
In-vacuum stepping motor
2-phase synchronous motor

Pulse Rate - Torque Characteristics

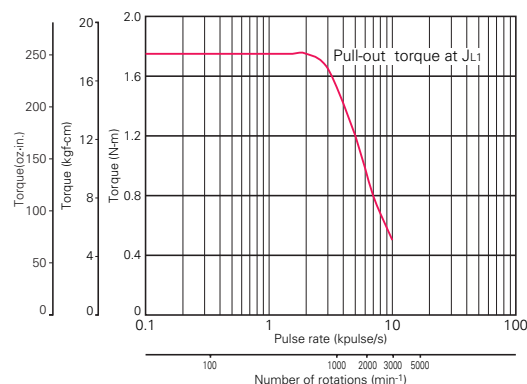
● 103H8221-0441



Sanyo constant current circuit

Source voltage: AC100V Operating current: 2A/phase, 2-phase energization (full-step)
J_{L1}=[7.4x10⁻⁴kg·m² (40.46 oz-in²) Use the rubber coupling]

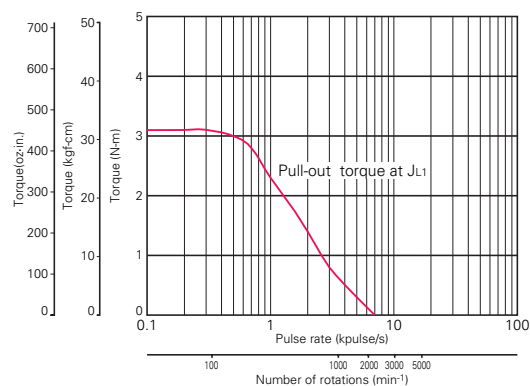
● 103H8221-0941



Sanyo constant current circuit

Source voltage: AC100V Operating current: 4A/phase, 2-phase energization (full-step)
J_{L1}=[7.4x10⁻⁴kg·m² (40.46 oz-in²) Use the rubber coupling]

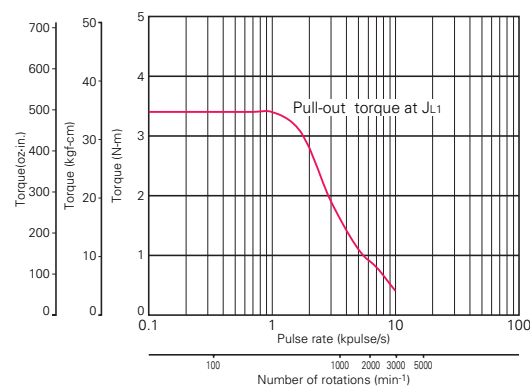
● 103H8222-0441



Sanyo constant current circuit

Source voltage: AC100V Operating current: 2A/phase, 2-phase energization (full-step)
J_{L1}=[15.3x10⁻⁴kg·m² (83.65 oz-in²) Use the rubber coupling]

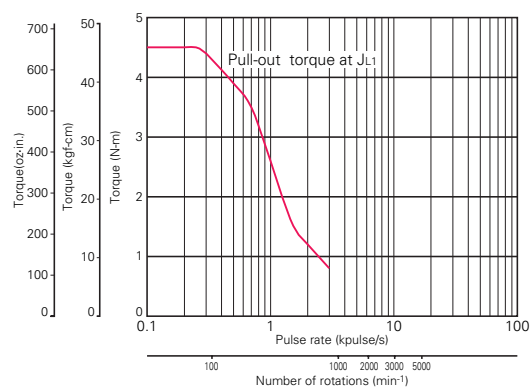
● 103H8222-0941



Sanyo constant current circuit

Source voltage: AC100V Operating current: 4A/phase, 2-phase energization (full-step)
J_{L1}=[15.3x10⁻⁴kg·m² (83.65 oz-in²) Use the rubber coupling]

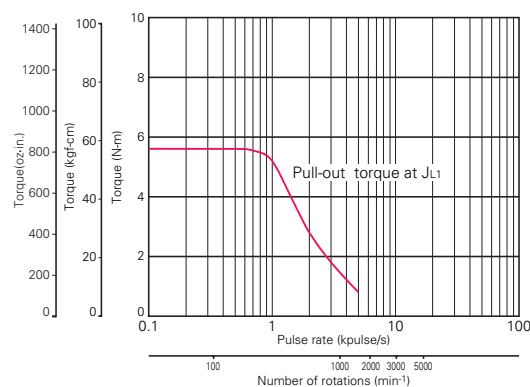
● 103H8223-0441



Sanyo constant current circuit

Source voltage: AC100V Operating current: 2A/phase, 2-phase energization (full-step)
J_{L1}=[15.3x10⁻⁴kg·m² (83.65 oz-in²) Use the rubber coupling]

● 103H8223-0941

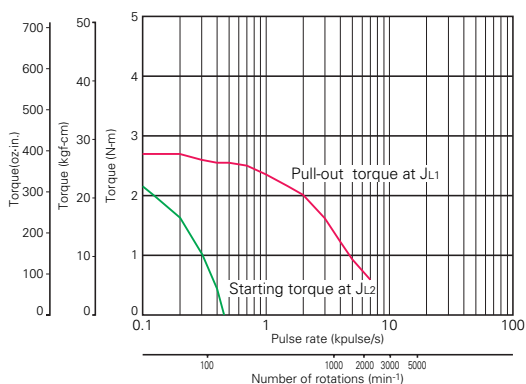


Sanyo constant current circuit

Source voltage: AC100V Operating current: 4A/phase, 2-phase energization (full-step)
J_{L1}=[15.3x10⁻⁴kg·m² (83.65 oz-in²) Use the rubber coupling]

Pulse Rate - Torque Characteristics

● 103H8221-5041

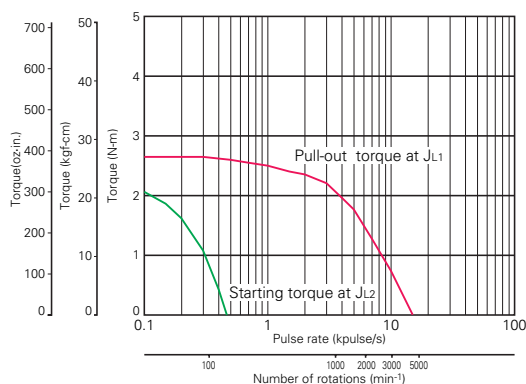


Sanyo constant current circuit

Source voltage: AC100V Operating current: 2A/phase, 2-phase energization (full-step)

J_{L1}=[7.4x10⁻⁴kg·m² (40.46 oz-in²) Use the rubber coupling]J_{L2}=[7.4x10⁻⁴kg·m² (40.46 oz-in²) Use the direct coupling]

● 103H8221-5141

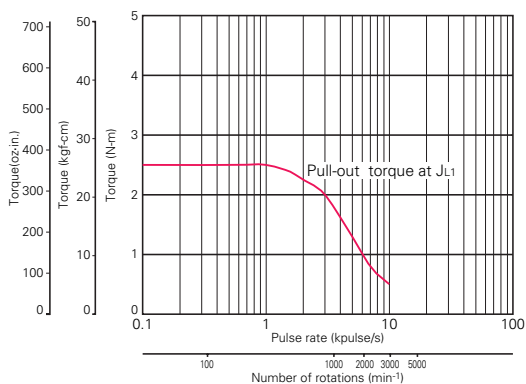


Sanyo constant current circuit

Source voltage: AC100V Operating current: 4A/phase, 2-phase energization (full-step)

J_{L1}=[7.4x10⁻⁴kg·m² (40.46 oz-in²) Use the rubber coupling]J_{L2}=[7.4x10⁻⁴kg·m² (40.46 oz-in²) Use the direct coupling]

● 103H8221-5241

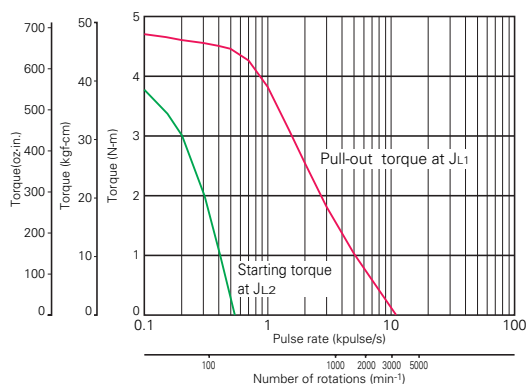


Sanyo constant current circuit

Source voltage: AC100V Operating current: 6A/phase, 2-phase energization (full-step)

J_{L1}=[15.3x10⁻⁴kg·m² (83.65 oz-in²) Use the rubber coupling]J_{L2}=[15.3x10⁻⁴kg·m² (83.65 oz-in²) Use the direct coupling]

● 103H8222-5041

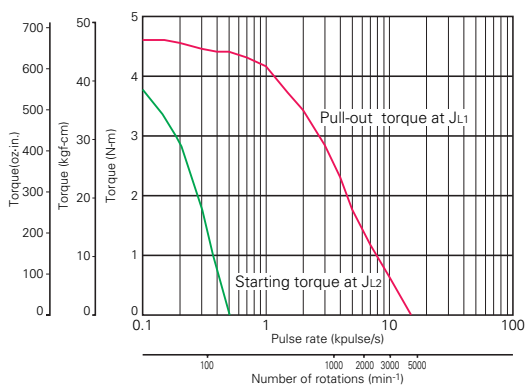


Sanyo constant current circuit

Source voltage: AC100V Operating current: 2A/phase, 2-phase energization (full-step)

J_{L1}=[15.3x10⁻⁴kg·m² (83.65 oz-in²) Use the rubber coupling]J_{L2}=[15.3x10⁻⁴kg·m² (83.65 oz-in²) Use the direct coupling]

● 103H8222-5141

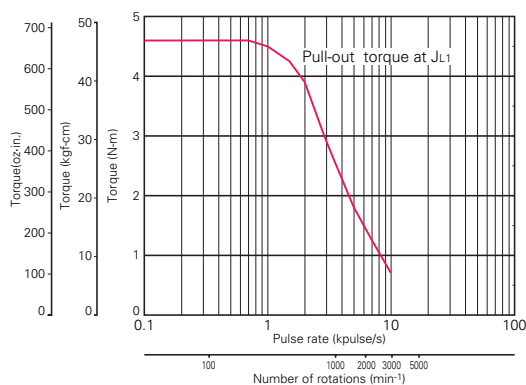


Sanyo constant current circuit

Source voltage: AC100V Operating current: 4A/phase, 2-phase energization (full-step)

J_{L1}=[15.3x10⁻⁴kg·m² (83.65 oz-in²) Use the rubber coupling]J_{L2}=[15.3x10⁻⁴kg·m² (83.65 oz-in²) Use the direct coupling]

● 103H8222-5241



Sanyo constant current circuit

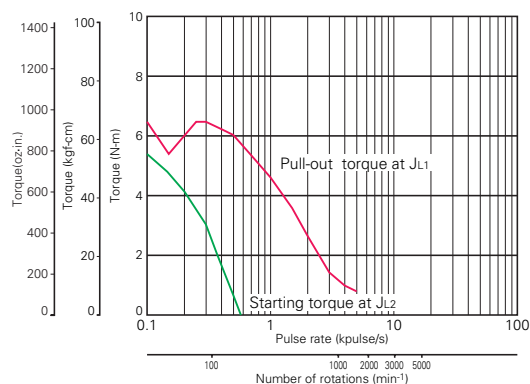
Source voltage: AC100V Operating current: 6A/phase, 2-phase energization (full-step)

J_{L1}=[15.3x10⁻⁴kg·m² (83.65 oz-in²) Use the rubber coupling]

35mm(1.38)/1.8"
 38mm(1.54)/0.9"
 42mm(1.65)/0.9"
 28mm(1.10)/1.8"
 42mm(1.65)/1.8"
 50mm(1.97)/1.8"
 56mm(2.20)/1.8"
 60mm(2.36)/1.8"
 86mm(3.39)/1.8"
 106mm(4.17)/1.8"
 56mm(2.20)/CE
 86mm(3.39)/CE
 106mm(4.17)/CE
 Specifications of 2-phase stepping motor
 In-vacuum stepping motor
 2-phase synchronous motor

Pulse Rate - Torque Characteristics

● 103H8223-5041



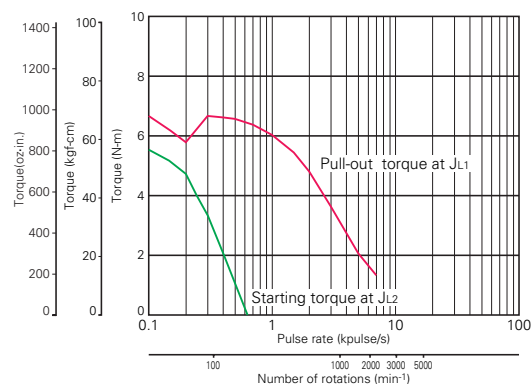
Sanyo constant current circuit

Source voltage: AC100V Operating current: 2A/phase, 2-phase energization (full-step)

J_{L1}=[15.3x10⁻⁴kg·m² (83.65 oz-in²) Use the rubber coupling]

J_{L2}=[15.3x10⁻⁴kg·m² (83.65 oz-in²) Use the direct coupling]

● 103H8223-5141



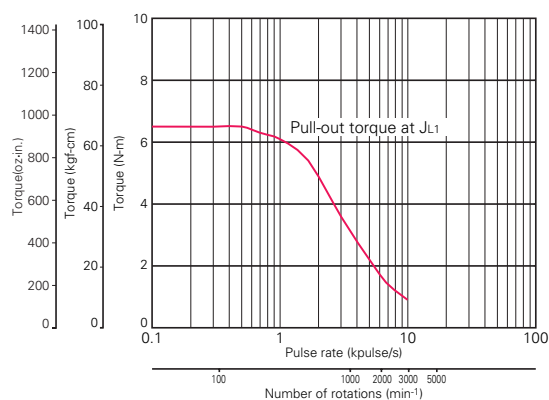
Sanyo constant current circuit

Source voltage: AC100V Operating current: 4A/phase, 2-phase energization (full-step)

J_{L1}=[15.3x10⁻⁴kg·m² (83.65 oz-in²) Use the rubber coupling]

J_{L2}=[15.3x10⁻⁴kg·m² (83.65 oz-in²) Use the direct coupling]

● 103H8223-5241



Sanyo constant current circuit

Source voltage: AC100V Operating current: 6A/phase, 2-phase energization (full-step)

J_{L1}=[15.3x10⁻⁴kg·m² (83.65 oz-in²) Use the rubber coupling]